

CENTRO DE ENSEÑANZA TÉCNICA Y SUPERIOR



Escuela de Ingeniería

Machine Learning

Exercises Unit 1- Bayesian Curl

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1.

EXERCISE 1 - IDENTIFY THE COMPONENTS



Your phone's fingerprint unlock:

$$P(\text{owner} \mid \text{match}) = \frac{P(\text{match} \mid \text{owner}) P(\text{owner})}{P(\text{match})}$$

Identify: prior, likelihood, posterior, evidence.

Then argue: why is $P(\text{owner})$ very high for a phone in your pocket, and how does that change the conclusion compared to the IDS example?

(5 minutes)

Equation Components

- Prior: $P(\text{owner})$ — The initial probability that the person attempting to unlock the device is the true owner, before the fingerprint is even evaluated.
- Likelihood: $P(\text{match} \mid \text{owner})$ — The probability of a successful sensor match given that the person is actually the owner (the true positive rate).
- Posterior: $P(\text{owner} \mid \text{match})$ — The updated probability that the person is the owner, given the new evidence of a successful fingerprint match.
- Evidence: $P(\text{match})$ — The overall, total probability of a match occurring on the sensor under any circumstance, regardless of who is using it.

Contextual Impact on Probability

The value of $P(\text{owner})$ is exceptionally high for a phone in your pocket because you maintain physical custody of the device. The vast majority of unlock attempts in this specific scenario are performed by you, pushing the base probability (prior) close to 100%.

This drastically shifts the statistical conclusion compared to an Intrusion Detection System (IDS). In a network environment, the prior probability of an actual cyberattack is extremely low because normal, legitimate traffic massively outnumbers malicious traffic. This imbalance leads to the "false positive paradox," where even a highly accurate IDS generates numerous false alarms because the base rate of attacks is so small. Conversely, with the phone in your pocket, the dominant prior probability, $P(\text{owner})$, means a positive fingerprint match simply confirms an already near-certain event, making the final posterior probability, $P(\text{owner} \mid \text{match})$, incredibly reliable.

2.

EXERCISE 2 - UPDATE A BETA PRIOR



A second department runs the same drill. Same industry prior, $\text{Beta}(\alpha_0 = 2, \beta_0 = 8)$. This time **10 employees** get the email and **4 click**.

a) Write the posterior. b) Compute its mean. c) Compare it with the MLE and the prior mean.

Check: the posterior mean lands exactly halfway between the two. Why? Count the real employees against the pretend ones.

(10 minutes)

a) Write the posterior

- Prior parameters: $\text{Alpha}_0 = 2$, $\text{Beta}_0 = 8$
- Observed data: 10 total employees (trials), 4 clicked (successes), 6 did not click (failures).
- Conjugacy update rule:
 - $\text{Alpha}_1 = \text{Alpha}_0 + \text{successes} = 2 + 4 = 6$
 - $\text{Beta}_1 = \text{Beta}_0 + \text{failures} = 8 + 6 = 14$
- Posterior Distribution: $\text{Beta}(6, 14)$

b) Compute its mean

- Formula: $\text{Mean} = \text{Alpha} / (\text{Alpha} + \text{Beta})$
- Posterior Mean: $6 / (6 + 14) = 6 / 20 = 0.30$ (or 30%)

c) Compare it with the MLE and the prior mean

- Prior Mean: $2 / (2 + 8) = 2 / 10 = 0.20$ (or 20%)
- Maximum Likelihood Estimate (MLE): $4 / 10 = 0.40$ (or 40%)
- Comparison: The posterior mean (0.30) lies directly at the midpoint between the prior mean (0.20) and the MLE (0.40).

Why does the posterior mean land exactly halfway? The sum of the prior parameters ($\text{Alpha}_0 + \text{Beta}_0 = 10$) represents 10 "pretend" employees worth of prior weight. The actual experiment observed 10 "real" employees. Because the weight of the prior evidence (10 pretend observations) is identical to the sample size of the new data (10 real observations), both sources of information carry equal weight (50% each), placing the posterior mean exactly halfway between them.

3.

EXERCISE 3 - HOW STRONG IS THE PRIOR? (CODING)



Same drill data every time (5 clicks of 20). Four priors, **all with mean 0.2**, differing only in strength:

```
clicks, ignores = 5, 15
for a0, b0 in [(1, 4), (2, 8), (10, 40), (100, 400)]:
    a, b = a0 + clicks, b0 + ignores
    print(f"prior Beta({a0},{b0}) with prior weight {a0+b0:>3} "
          f"-> posterior mean {a/(a+b):.3f}")
```

- Which prior gives an answer closest to the MLE (0.250)? Which is closest to the prior mean (0.200)?
- State the rule in one sentence, in terms of prior weight versus real observations.

(15 minutes coding)

40

```
clicks, ignores = 5, 15

for a0, b0 in [(1, 4), (2, 8), (10, 40), (100, 400)]:
    a, b = a0 + clicks, b0 + ignores
    print(f"prior Beta({a0},{b0}) with prior weight {a0+b0:>3} "
          f"-> posterior mean {a/(a+b):.3f}")
```

[1] ✓ 0.0s Python

```
... prior Beta(1,4) with prior weight 5 -> posterior mean 0.240
prior Beta(2,8) with prior weight 10 -> posterior mean 0.233
prior Beta(10,40) with prior weight 50 -> posterior mean 0.214
prior Beta(100,400) with prior weight 500 -> posterior mean 0.202
```

Question 1: Which prior gives an answer closest to the MLE (0.250)? Which is closest to the prior mean (0.200)?

- Closest to the MLE (0.250): The Beta(1, 4) prior, resulting in a posterior mean of 0.240.
- Closest to the prior mean (0.200): The Beta(100, 400) prior, resulting in a posterior mean of 0.202.

Question 2: State the rule in one sentence, in terms of prior weight versus real observations. When the prior weight is small relative to the number of real observations, the empirical data dominates and the posterior stays close to the MLE; conversely, when the prior weight is much larger than the sample size, the prior dominates and pulls the posterior mean toward the prior mean.